AI/ML

for

Sustainable Infrastructure Systems

(CE60258)

One Month Integrated Big Problem
on
Post-Tensioned Prestressed Concrete
I-Girder Bridge

October, 2026



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Indian Institute of Technology Kharagpur

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October, 2026

Course Code:	CE60258
Credits:	3–0–0 (3 Credits)
Announcement of Big Problem:	September 09, 2026
Clarification:	October 07, 2026
Clarification (if required)	October 14, 2026
Submission:	October 31, 2026

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Before You Begin

This problem has been chosen not merely to test what you already know, but to explore your **curiosity, willingness to venture into an unfamiliar problem, and ability to learn while solving it**.

You may not know at the beginning how to solve every part of the problem. That is intentional.

An engineer will often encounter a problem for which the complete solution is not known beforehand. The important question is whether one is willing to understand the problem, explore possible approaches, ask questions, learn what is necessary, and gradually develop a solution.

In the age of AI, you are not alone in this journey. You may take the assistance of **any AI chatbot of your choice**. You may ask it to explain concepts, suggest approaches, perform calculations, develop programs, generate synthetic data, analyse results, or help you explore alternative solutions.

However, the purpose of this problem is **not** to give a prompt to an AI chatbot, receive a response, and submit that response as your solution.

You are expected to interact with AI. Ask questions. Examine its responses. Question its assumptions. Ask why. Ask for alternatives. Check calculations. Modify the approach. Develop programs when useful. Test the results. Return to the AI when something does not appear reasonable. Continue this interaction until the solution becomes **your solution**.

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One-Month Integrated Big Problem on Post-Tensioned Prestressed Concrete I-Girder Bridge

Course: AI/ML for Sustainable Infrastructure Systems
Duration: One Month
Mode: Individual Integrated Engineering Problem
AI Assistance: Any AI chatbot may be used.

1.1 Objective

The objective of this integrated problem is to study a **post-tensioned prestressed concrete I-girder bridge** as an infrastructure system throughout its service life.

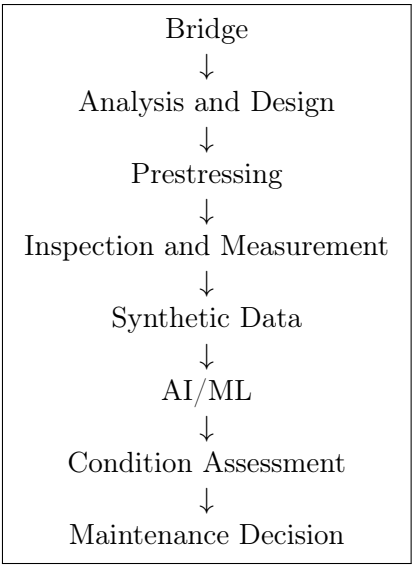
The work shall be carried out individually. Each student shall select one span only and complete the entire integrated study independently.

The problem integrates

- structural analysis and design,
- prestressing,
- visual inspection,
- non-destructive testing,
- displacement and strain measurement,
- vibration measurement and signal processing,
- preparation of synthetic data,
- feature extraction,
- AI/ML-assisted condition assessment, and
- maintenance planning.

The overall philosophy of the problem may be represented

as



1.2 Bridge Configuration

Consider a simply supported **post-tensioned prestressed concrete I-girder bridge**.

The bridge shall have

- three traffic lanes,
- footpaths on both sides,
- crash barriers on both sides,
- longitudinal post-tensioned PSC I-girders,
- RCC deck slab,
- cross-girders or diaphragms, as appropriate,
- bearings, and
- expansion joints.

Each student shall independently select **one span only** from the following alternatives:

$$\boxed{L = 30 \text{ m}, \qquad L = 33 \text{ m}, \qquad L = 36 \text{ m}}$$

For the selected span, each student shall independently

develop a suitable bridge configuration.

The student shall decide, with appropriate engineering justification,

- overall bridge cross-section,
- carriageway and footpath arrangement,
- number of longitudinal I-girders,
- spacing of longitudinal girders,
- dimensions of the PSC I-girders,
- deck slab thickness,
- arrangement of cross-girders or diaphragms, and
- other relevant structural dimensions.

Different students selecting the same span are not expected to arrive at identical bridge configurations. The adopted configuration and dimensions shall be technically reasonable and properly justified.

1.3 Structural Analysis and Preliminary Design

The bridge shall first be studied in its assumed healthy condition.

Students shall identify the important loads and develop an appropriate structural model of the bridge.

Important structural response quantities may include

$$M, \qquad V, \qquad \sigma, \qquad \varepsilon, \qquad \delta.$$

The analysis may be performed using

- analytical calculations,
- Python programs,
- suitable structural analysis software,
- AI-generated computational programs, or
- an appropriate combination of these methods.

Students are encouraged to use **Vibe Coding** for developing computational tools for analysis, parametric studies, data processing, visualisation, and report preparation.

1.4 Prestressing System

For the adopted bridge configuration, each student shall independently develop a suitable post-tensioning system.

The student shall determine

- number of prestressing cables,
- number and arrangement of strands or tendons in each cable,
- cable profile,
- initial prestressing force,
- prestressing losses,
- effective prestressing force, and
- stresses at important stages.

The initial prestressing force may be represented by

$$P_0,$$

whereas the effective prestressing force after losses may be represented by

$$P_e.$$

Thus,

$$\boxed{P_e < P_0}$$

The effect of prestressing on the stresses in the girder at important sections shall be examined.

AI assistance may be used for preliminary calculations and development of computational programs. The adopted prestressing arrangement and calculated results, however, shall be examined and verified from an engineering point of view.

There is no requirement that students selecting the same span obtain the same number of cables, cable arrangement, or prestressing force. The proposed solution shall, however, be technically reasonable and properly justified.

1.5 Healthy Baseline Condition

Before introducing deterioration, students shall establish a **baseline condition** representing the healthy bridge.

The baseline may include quantities such as

- effective prestressing force,
- stresses in the girder,
- displacement,
- strain,
- natural frequencies,
- acceleration response, and
- relevant NDT parameters.

The baseline condition may be represented as

Healthy Bridge $\longrightarrow$ Baseline Structural and Monitoring Data

Subsequent observations and measurements shall be compared with this baseline.

1.6 Visual Inspection

A systematic visual inspection scheme shall be developed for the bridge.

The inspection may include components such as

- deck slab,
- longitudinal PSC I-girders,
- cross-girders or diaphragms,
- footpaths,
- crash barriers,
- bearings,
- expansion joints,
- drainage arrangements,
- pier caps,
- piers, and
- abutments.

Possible observations may include

Cracking	Spalling	Corrosion Staining
Leakage	Honeycombing	Surface Deterioration

together with other relevant signs of distress.
The inspection process should identify

Bridge → Span → Component → Location → Observation

Students may develop a digital visual inspection application using Python or a web-based implementation.

1.7 Non-Destructive Testing

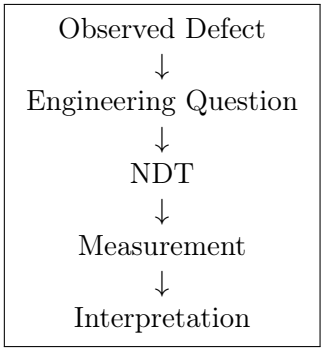
Based on the visual inspection and assumed condition of the bridge, an appropriate non-destructive testing programme shall be developed.

Suitable tests may include

- Rebound Hammer,
- Ultrasonic Pulse Velocity,
- Cover Meter,
- corrosion assessment, and
- other appropriate non-destructive tests.

The selection of an NDT method should be related to a specific engineering question.

The procedure may therefore be represented as



Students shall identify suitable locations for testing and explain the engineering significance of the measurements.

1.8 Structural Response Measurement

Suitable structural response quantities shall be selected for monitoring the bridge.

These may include

Displacement	Strain	Acceleration

Students shall propose

- quantities to be measured,
- suitable sensors,
- sensor locations,
- measurement frequency or sampling frequency, where relevant,
- duration of measurement, and
- interpretation of the measured response.

1.9 Vibration Measurement

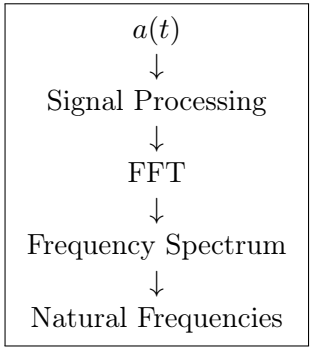
Accelerometers shall be considered at suitable locations on the bridge for measurement of its vibration response.

Students shall propose

- number of accelerometers,
- locations of accelerometers,

- direction of measurement,
- sampling frequency,
- duration of measurement, and
- possible sources of excitation.

A typical vibration measurement and signal-processing sequence is



The vibration characteristics of the healthy bridge shall form part of the baseline condition.

1.10 Preparation of Synthetic Data

Students shall prepare representative **synthetic inspection and monitoring data** for the bridge.

The purpose of synthetic data is to simulate realistic changes in the measured quantities under different assumed conditions of the bridge.

Synthetic data may be prepared for

- visual inspection,
- displacement,
- strain,
- NDT measurements,
- acceleration time histories,
- natural frequencies,
- prestressing force, and
- other relevant structural quantities.

Example: Synthetic Displacement Data

An illustrative set of synthetic displacement data may be

Condition	**Mid-span Displacement (mm)**
Healthy	12.4
Condition 1	12.9
Condition 2	13.8
Condition 3	15.1

These values are illustrative only. Students shall prepare suitable synthetic data based on their adopted structural model and assumptions.

Example: Synthetic NDT Data

A representative synthetic NDT dataset may be

Location	**Rebound Number**	**UPV (km/s)**	**Assumed Condition**
G1-1	34	4.25	Healthy
G1-2	32	4.05	Minor Change
G1-3	27	3.55	Deteriorated
G1-4	23	3.10	Severe

The above values are intended only as examples of synthetic data and shall not be interpreted as prescribed condition limits.

Example: Synthetic Vibration Data

Suppose the first three natural frequencies of the healthy bridge are

$$f_1 = 3.20 \text{ Hz}, \qquad f_2 = 5.85 \text{ Hz}, \qquad f_3 = 8.40 \text{ Hz}.$$

An assumed deteriorated condition may produce

$$f_1 = 3.05 \text{ Hz}, \qquad f_2 = 5.62 \text{ Hz}, \qquad f_3 = 8.12 \text{ Hz}.$$

Students shall investigate the possible engineering significance of such changes.

A change in frequency should not automatically be interpreted as a particular type of damage. Other factors affecting structural response should also be considered.

1.11 Deterioration

After establishing the healthy baseline condition, each student shall develop suitable deterioration scenarios for the bridge.

Possible deterioration mechanisms may include

- loss of prestressing force,
- concrete deterioration,
- cracking,
- corrosion-related deterioration,
- bearing deterioration,
- local reduction in stiffness, and
- other suitable deterioration mechanisms.

The student shall decide the type, location, extent or magnitude of the deterioration and its expected influence on the structural properties and measured response. The deterioration values shall not be prescribed. They shall be selected by the student on a physically reasonable basis and appropriately justified.

Students shall examine how deterioration affects the quantities being measured or calculated.

1.12 Feature Extraction

Students shall identify useful features from the synthetic inspection, NDT, and monitoring data.

For vibration measurements, possible features may include

$$f_1, \qquad f_2, \qquad f_3, \qquad a_{\mathrm{RMS}}, \qquad a_{\max}.$$

For inspection and NDT data, possible features may include

Crack Width, Rebound Number, UPV, Condition Score.

For structural response, additional features may include

$$\delta_{\max}, \qquad \varepsilon_{\max}, \qquad P_e.$$

A possible feature vector may therefore be written as

$$\boxed{\mathbf{x} = [f_1,\ f_2,\ a_{\mathrm{RMS}},\ \delta_{\max},\ \varepsilon_{\max},\ \text{UPV},\ \text{Rebound Number}]}$$

Students are not required to use this particular feature vector. Selection of meaningful features forms part of the problem.

1.13 AI/ML-Assisted Condition Assessment

The synthetic data shall be used to investigate how AI/ML may assist in the assessment of the bridge.

Depending on the nature and quantity of the generated data, students may explore

- regression,
- classification,
- clustering,
- anomaly detection,
- trend identification,
- feature-based condition assessment,
- vibration-based assessment, and
- data fusion.

The purpose is not merely to obtain a high numerical accuracy from an AI/ML model.

Students shall explain

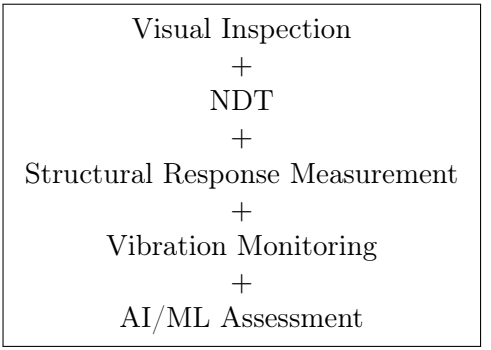
- why particular features have been selected,
- what physical meaning the features have,
- what the AI/ML output indicates,
- whether the result is physically reasonable, and
- how the result may assist an engineer in assessing the

bridge.

1.14 Maintenance Scheme

Students shall finally develop a maintenance scheme for the bridge based on the integrated assessment.

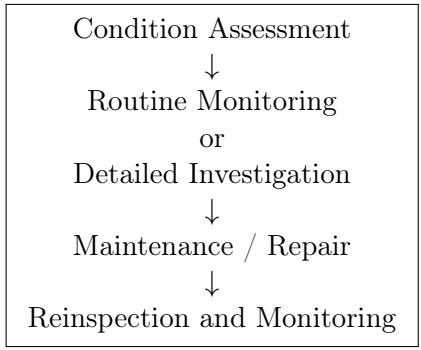
The maintenance decision should consider information obtained from



The maintenance strategy may include

- routine inspection and monitoring,
- detailed inspection,
- additional NDT,
- additional instrumentation,
- preventive maintenance,
- repair,
- detailed structural assessment, and
- subsequent inspection and monitoring.

A possible decision sequence is



Students shall justify the proposed maintenance actions on the basis of the available engineering information.

1.15 Use of AI Chatbots

Students may use **any AI chatbot** during the development of the integrated problem.

AI assistance may be used for

- understanding and formulation of the problem,
- preliminary structural analysis and design,
- prestressing calculations,
- development of Python programs,
- Vibe Coding,
- preparation of synthetic data,
- signal processing,
- feature extraction,
- development of AI/ML models,
- visualisation of results,
- development of inspection applications, and
- preparation and improvement of the report.

However,

$$\text{AI-Generated Result} \neq \text{Verified Engineering Result}$$

Students remain responsible for

- assumptions,
- engineering models,
- equations,
- units,
- computational results,
- physical interpretation, and
- engineering decisions.

1.16 Individual Nature of the Problem

This is an **individual one-month integrated problem**. It is not a group assignment.

Each student shall select one span only and shall be responsible for the complete development of that bridge, including

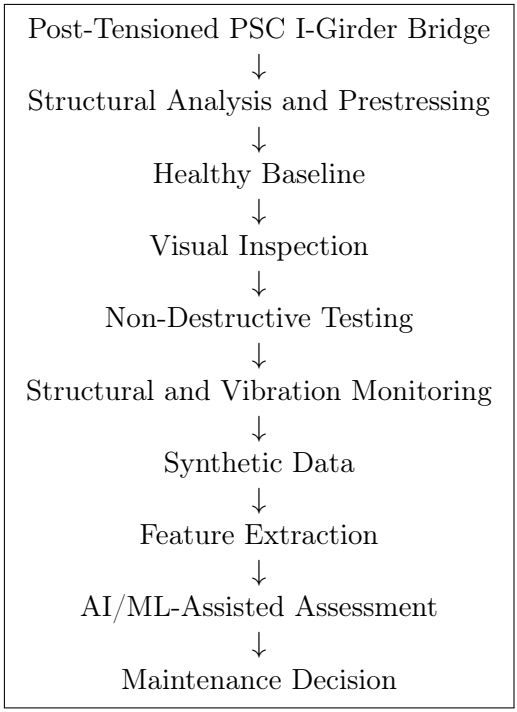
- bridge configuration and dimensions,
- structural analysis,
- prestressing system and prestressing forces,
- healthy baseline condition,
- visual inspection and NDT scheme,
- structural and vibration monitoring scheme,
- deterioration scenarios and deterioration values,
- synthetic data,
- feature extraction and AI/ML application, and
- maintenance scheme.

Different students, including those selecting the same span, may obtain different technically acceptable solutions. Such differences are expected and form an important part of the problem.

1.17 Expected Final Outcome

The final submission should present the study as the **integrated story of one bridge**, rather than as a collection of independent exercises.

The complete study may be represented as


Post-Tensioned PSC I-Girder Bridge
Structural Analysis and Prestressing
Healthy Baseline
Visual Inspection
Non-Destructive Testing
Structural and Vibration Monitoring
Synthetic Data
Feature Extraction
AI/ML-Assisted Assessment
Maintenance Decision

1.18 Evaluation

The weightage of marks for the **One-Month Integrated Big Problem** shall be **10 marks**, as mentioned in the *Course Information, Evaluation and Grading*.

For completeness, the subsection related to the award of certificates is reproduced below from the *Course Information, Evaluation and Grading*.

1.18.1 Certificate for the Integrated One-Month Big Problem

Students who successfully complete the Integrated One-Month Big Problem will receive a separate certificate in addition to the course grade.

The certificates will be awarded according to the criteria given in Table 1.1. Certificates will be issued only upon request after the publication of the course grades.

Table 1.1: Certificate Criteria

Project Score	**Certificate**
8.0 and above	Commendable Certificate
6.0 to below 8.0	Appreciation Certificate
Below 6.0	No Certificate

The certificate recognises the quality of the student's work on the Integrated One-Month Big Problem and is awarded independently of the final course grade.

